

Cinematography For Stylized Motion Pictures

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Figure 1: Split screen of photoreal render from RenderMan with Stylized approach containing illustrative lighting and shadows based off photoreal data

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1 INTRODUCTION

One of the biggest look development and visual design changes in recent years on feature film animations has been the use of “stylized rendering” - painterly, watercolor, illustrative lighting, and linework techniques – in cinematography. These are used in lighting design, shadow placement, focus & depth of field through various animated R&D and feature film projects.

2 TRADITIONAL CINEMATOGRAPHY

Traditional cinematography techniques are looked at in how Depth Of Field is used to drive the viewers eye for focus and storytelling.

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Additionally, lighting design and shadow placement drive elements of story, focus, mood, and character. These techniques have been perfected over the many years of cinema and are successful tools in storytelling.

3 LEVEL OF STYLE

We are enhancing traditional cinematography techniques with the concept and practice of Level Of Style: where each level has a unique and deliberate design, instead of being out of focus.

For example, different designs of Levels Of Style could be broad brush strokes instead of photoreal detail, abstract linework, painterly treatments, illustrative shadows, or hatching techniques.

These can be broken down in a Level of Style Guide for the project such as:

- Level 1: Full detail
- Level 2: Same geometry detail with reduced color palette
- Level 3: Simpler geometry outline
- Level 4: No textures, only flat colours

Various concept artists work, such as Syd Mead, Art Fitzpatrick and Von Kauffman is used as inspiration for Level of Style.

Level Of Style doesn't have to be solely based on camera distance either, it can be designed along light and shadow variations, and story driven characters: a certain character can have a simplified Level Of Style and be in full focus on screen, but the story dictates their style.



Figure 2: Traditional depth of field compared with painterly techniques based on camera distance

The way that Level Of Style blends from foreground to background, light and shadow, and story driven transitions, has many different creative possibilities and technological challenges: does a background element that is “out of focus” by having a simplified color palette and geometric shape blend into the foreground by a simple cross fade? Or do certain colors fade in first at a camera distance, followed by more detailed geometry?

On the one hand these techniques create more work for the filmmaker and technical challenges as opposed to them being blurred out with traditional out of focus lens techniques, yet at the same time every single aspect has more deliberate purpose and design to it, making it a rich and compelling design.

4 CAMERA

The key is to not draw the viewers eye to “out of focus” elements in the camera, but rather to drive different areas of focus to convey the story based on illustrative inspiration and concept art. We compare traditional “bokeh” depth of field on story elements as they go further into the background, with various different stylized approaches.

These various Level Of Style techniques can also be combined together; such as having linework only on the extreme foreground of the frame, with photoreal focus on characters in the center, and painterly brush techniques in the far distance. This has the effect on the observer of studying the frame for all the interesting Level Of Style, while still being focused on the intended focus region to convey the story.

In practice, we look at implementations of these techniques in-render with Pixar’s RenderMan Stylized Looks workflow and focusing the viewers eye with shadow density hatching textures based on camera distance, and custom GPU accelerated Blink API inside Nuke compositing software with live demos of tuning the camera distance to place broad brush strokes that go off geometry, in combination with linework.

5 LIGHTING DESIGN

In addition to camera techniques, lighting design is an important aspect of traditional cinematography while providing creative opportunities and technical challenges in Stylized Rendering.

Where we have traditional shadow placement and lighting design, we explore stylized directions that take in physically based lighting models, but place illustrative techniques such as linework,



Figure 3: Linework based on extreme foreground camera distance, and watercolor shadow



Figure 4: Traditional bokeh depth of field in shot with different lighting scenarios, compared with painterly and line techniques based on camera distance

painterly brush, cross hatching, and watercolor treatments throughout the lighting response.

In our practical implementations of these techniques, every unique light source in the scene can be either photoreal, or stylized with line work, painterly, or illustrative. These lights can be mixed and matched in one render or scene to have the artists design of photoreal mixed with stylized treatments based off that light source, and layered and daisy chained together.

Particularly of interest is when the lighting model is swapped for a different one: for example a dusk lighting scenario to a night lit by lamplight. These two lighting scenarios have predictable results in photoreal rendering; however with our Level Of Style approaches and illustrative techniques based on lighting, we show how the color of brushes drastically can change with different lighting scenarios for very interesting effect, as well as linework that is based on the light signal, and cross hatching color based on lighting.

6 IMPLEMENTATION

Christos Obretenov has implemented these techniques in Pixar’s RenderMan’s Stylized Looks, Foundry’s Nuke compositing software, Open Shading Language (OSL) with Sony Pictures Imageworks feature film projects Mitchell’s vs Machines (Digi Pro 2021 paper), and Spider Man: Across the Spideverse (2023).

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